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SIMATS ENGINEERING
ASSESSMENT TOOL 2
Case study

COURSE NAME: Computer Networks

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO3: Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions

Assessment Tool Weightage: 40%

Code of Conduct:

I, Avinash.M (192521196) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



38
40

A

QUESTIONS:4

Total Marks:40

1. A multinational IT company uses a cloud-based video conferencing platform for daily client meetings. During peak business hours, employees experience frequent voice distortion, frozen video frames, and delayed communication. Network monitoring reports indicate an **8% packet loss**, **180 ms jitter**, and fluctuating bandwidth utilization. The IT administrator suspects that the issue is related to the transport protocol or application-level buffering strategy.

Questions

1. Analyze whether the problem is primarily caused by the **transport protocol (TCP/UDP)** or the **application layer**.
2. Justify your answer based on the communication requirements of real-time multimedia applications.
3. Recommend suitable protocol-level or application-level improvements to enhance conferencing quality.

2. A multinational software company hosts its web services across multiple continents. Employees frequently report long delays before the application homepage loads. Network analysis shows that the **average DNS lookup time is approximately 800 ms**, significantly affecting user experience. The organization plans to redesign its DNS infrastructure while ensuring continuous service availability during server failures.

aQuestions

1. Analyze the reasons behind the high DNS resolution time.
2. Propose a suitable DNS architecture using techniques such as **Anycast DNS**, **DNS pre-fetching**, and **TTL optimization** to reduce the lookup time below **50 ms**.
3. Evaluate the advantages and limitations of your proposed architecture in terms of performance, scalability, and availability.

3. A cloud service provider delivers applications to millions of users worldwide. During promotional events, users experience intermittent delays in accessing cloud-hosted services. Investigation reveals that DNS queries are taking nearly **800 ms** because all requests are directed to a single primary DNS server. The company wants a highly available DNS solution capable of handling global traffic efficiently.

Questions

4. Examine the impact of centralized DNS architecture on application performance.
5. Design a distributed DNS solution that minimizes lookup latency while maintaining high availability.
6. Analyze how **TTL values**, **Anycast routing**, and **DNS caching** influence system performance and fault tolerance.

4.A financial institution operates an electronic stock trading platform where thousands of buy and sell orders are submitted every second. During market opening hours, the **average order acknowledgment latency increases from 1 ms to 40 ms**, resulting in delayed trade confirmations and customer dissatisfaction. Network engineers observe increased TCP retransmissions, longer connection establishment times, and excessive buffering.

Questions

1. Analyze three possible TCP-related factors responsible for the latency increase, considering **flow control**, **Nagle's algorithm**, and **SYN queue limitations**.
2. Explain how each factor affects transaction processing in high-frequency systems
3. Recommend appropriate TCP configuration changes to reduce latency and improve transaction reliability.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning

Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

C03-AT₁

Case Study

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Reg No: 192521196

Course: Computer Networks

Code: CSA0707

Date: 8/25/08/2028

Q1) Video conferencing Performance Analysis.

Case understanding:

The multinational (IT) company is experienced voice distortion, frozen video frames, and delayed communication during peak business hours.

i) Analyze whether the problem is primarily caused by TCP/UDP.

* The problem is primarily related to network condition and the transport / application strategy, rather than being caused by UDP alone.

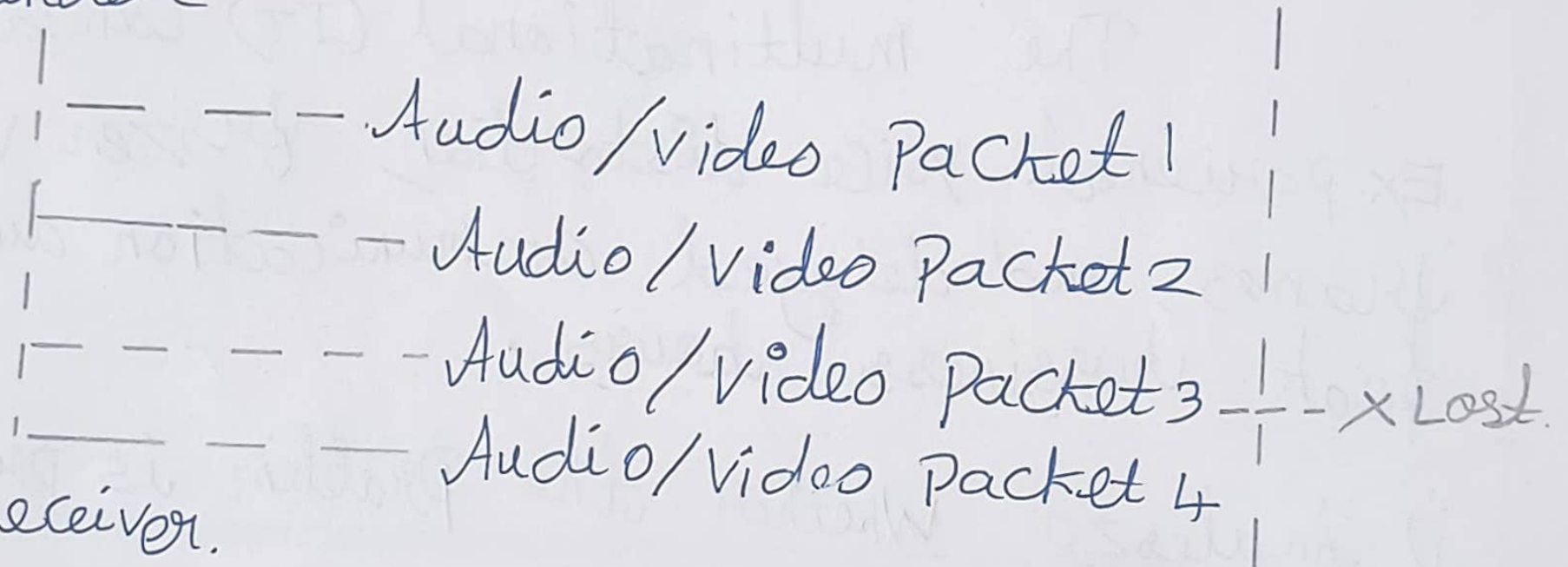
ii) Justification Based on Real-time multimedia Requirement:-

UDP Advantages:-

*> UDP does not require connection establishment acknowledgment or Retransmission of Every lost Packet.

*> Therefore, it can be Either delivery Packets with lower overhead and lower latency.

Sender



Receiver.

iii) Recommended improvements:-

A) use UDP - Based Real time Transport:
*> use UDP for real-time Audio/video traffic where appreciable with application - local mechanism to manage Packet loss and Jitter.

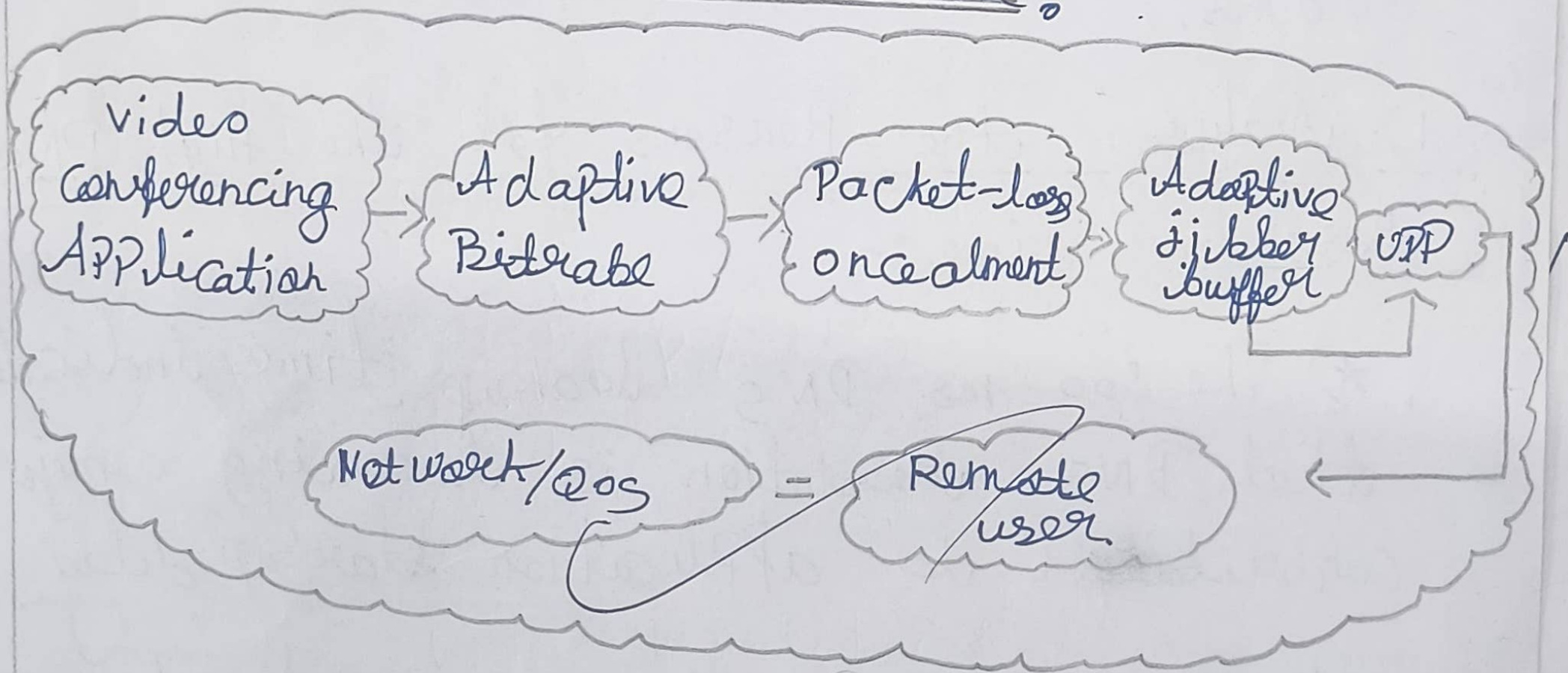
3) Adaptive Jitter buffer:

The conferencing applications should dynamically adjust its jitter buffer.

★) If jitter increases $\rightarrow$ Temporarily increase buffer size.

★) Avoid Excessively large buffer because increases.

iv) Recommended architecture:



v) Conclusion:

The best solution is a combined transport and application-level approach use a low-latency transport suitable for real-time media,

Q2) DNS infrastructure Redesign for a Multinational company or company.

Case understanding:-

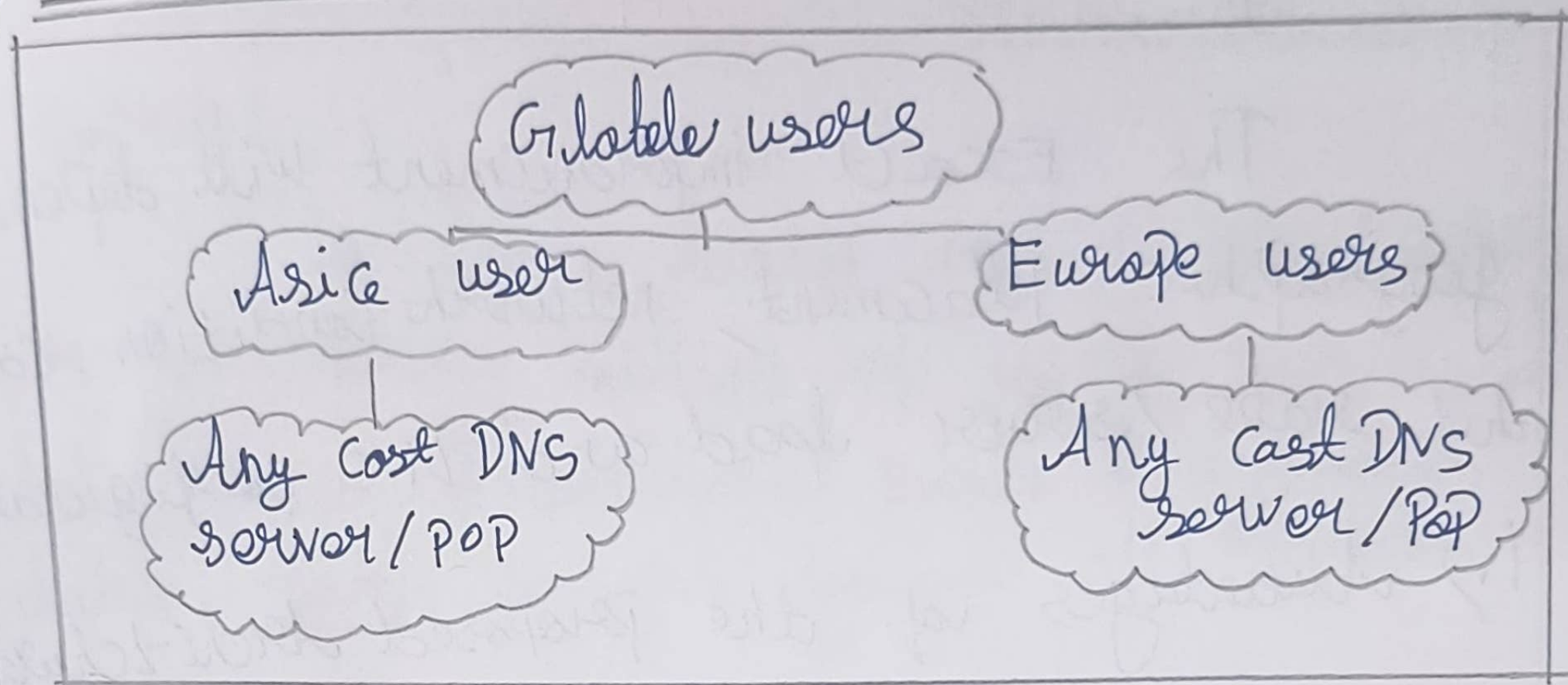
The multinational software company hosts web services across multiple continents. Employees experience long delay & before the application homepage loads because the average DNS lookup time is approximately 800 ms.

i) Analyze the Reasons for the High DNS Resolution time:-

* The 800 ms DNS lookup time indicates that DNS resolution is becoming a major contributor to application startup delay.

* The 800 ms DNS lookup can therefore significantly increase the time before the application begins loading.

1) Proposed DNS Architecture:-



A) Any cast DNS :-

Any cast DNS allows multiple DNS server in different geographic locations to advertise the same DNS service address. A user is normally routed to a suitable / nearby Anycast location.

B) DNS caching :-

DNS response should be cached at recursive resolvers and other associated network layers.

iii) Targeting DNS lookup Below 50ms.

The Exact improvement will depend on geographic Placement, network condition, cache hit rate / server load and DNS configuration.

iv) Advantages of the proposed Architecture:

- * Performance: - Any cast reduces geographic
- * Availability: If one DNS location becomes unavailable, traffic can be routed.

v) Limitation:

- * Require careful routing and monitoring
- * Cached records can become outdated.
- * May generated unnecessary queries.

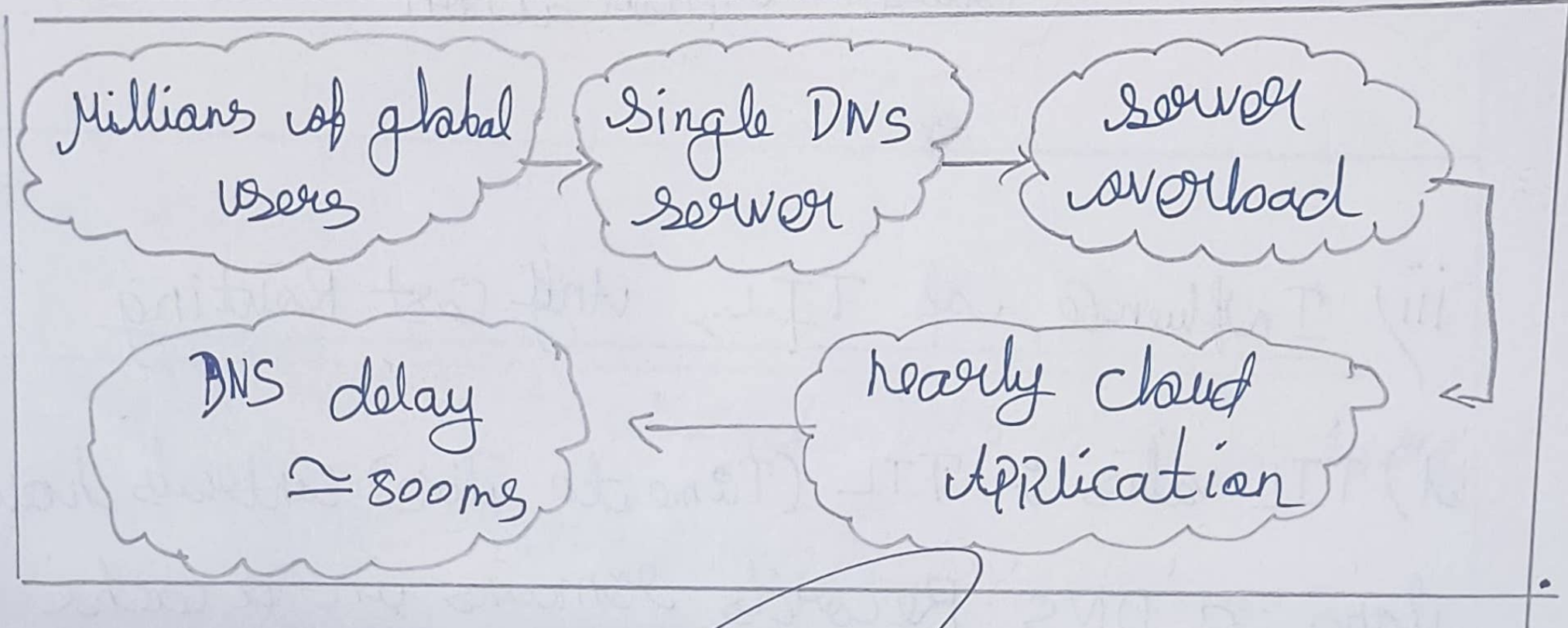
vi) Conclusion:

The 800 DNS lookup is primarily caused by inefficient DNS resolution infrastructure such as geographic distance.

Q3) Distributed DNS Solution for Global Cloud Source:-

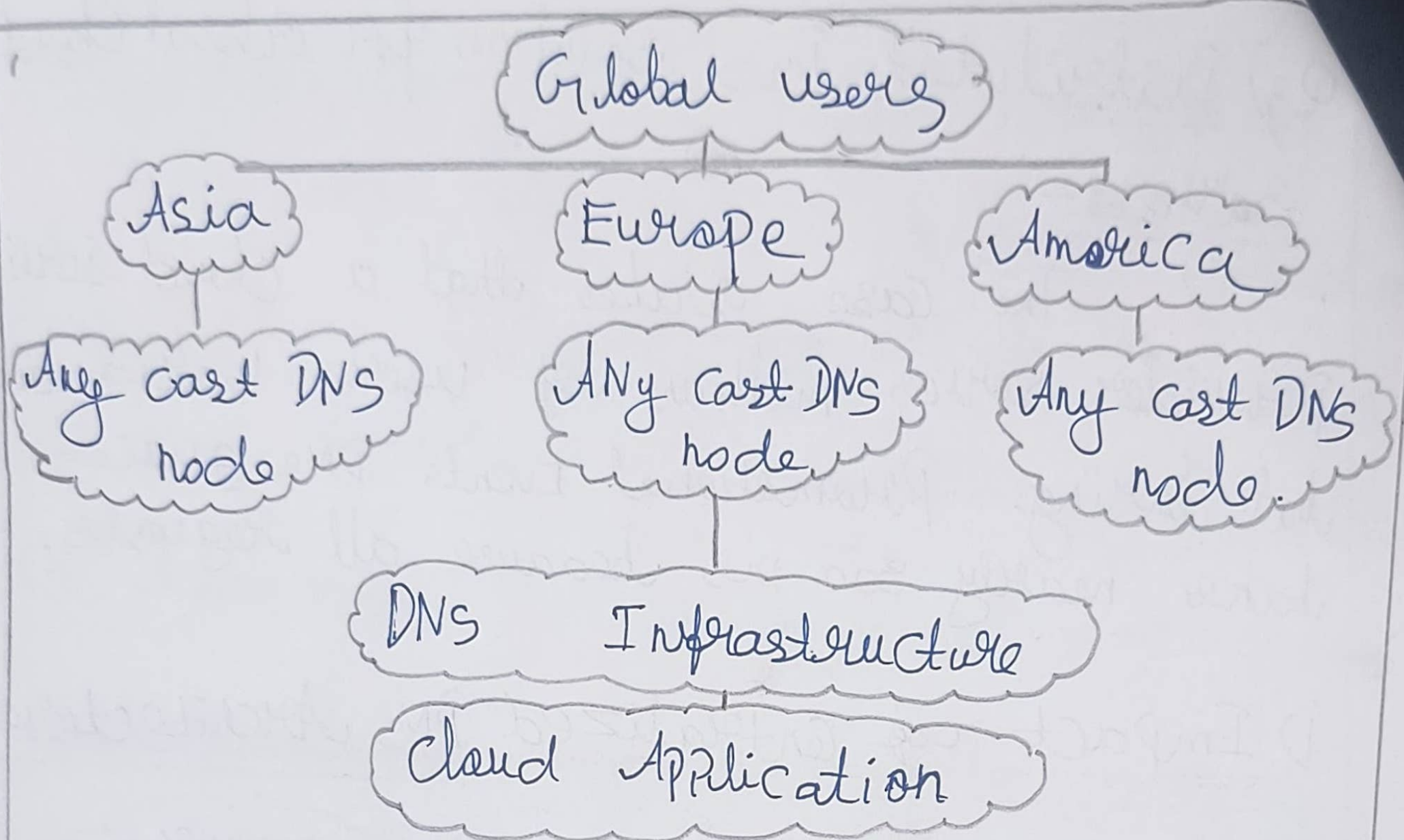
The case states that a cloud service provider serves millions of users worldwide but during promotional events DNS queries takes nearly 800 ms because all requests.

i) Impact of Centralized DNS Architecture



ii) Design of a Distributed DNS solution

The recommended solution is a distributed Anycast DNS architecture with DNS caching, appropriate values, and /multiple DNS location.



iii) Influence of TTL, Any Cast Routing

A) TTL values :- TTL (Time to live) controls how long a DNS Records remains in a cache

~~Appropriate~~ TTL :- A Properly selected TTL reduces unnecessary DNS queries while avoiding excessively stale records.

iv) Recommended Solution.

- * Any Cast DNS nodes across major
- * Multiple DNS server to eliminate the Single Point of failure.
- * DNS caching to repeated.
- * Continuous DNS monitoring to identify field nodes.

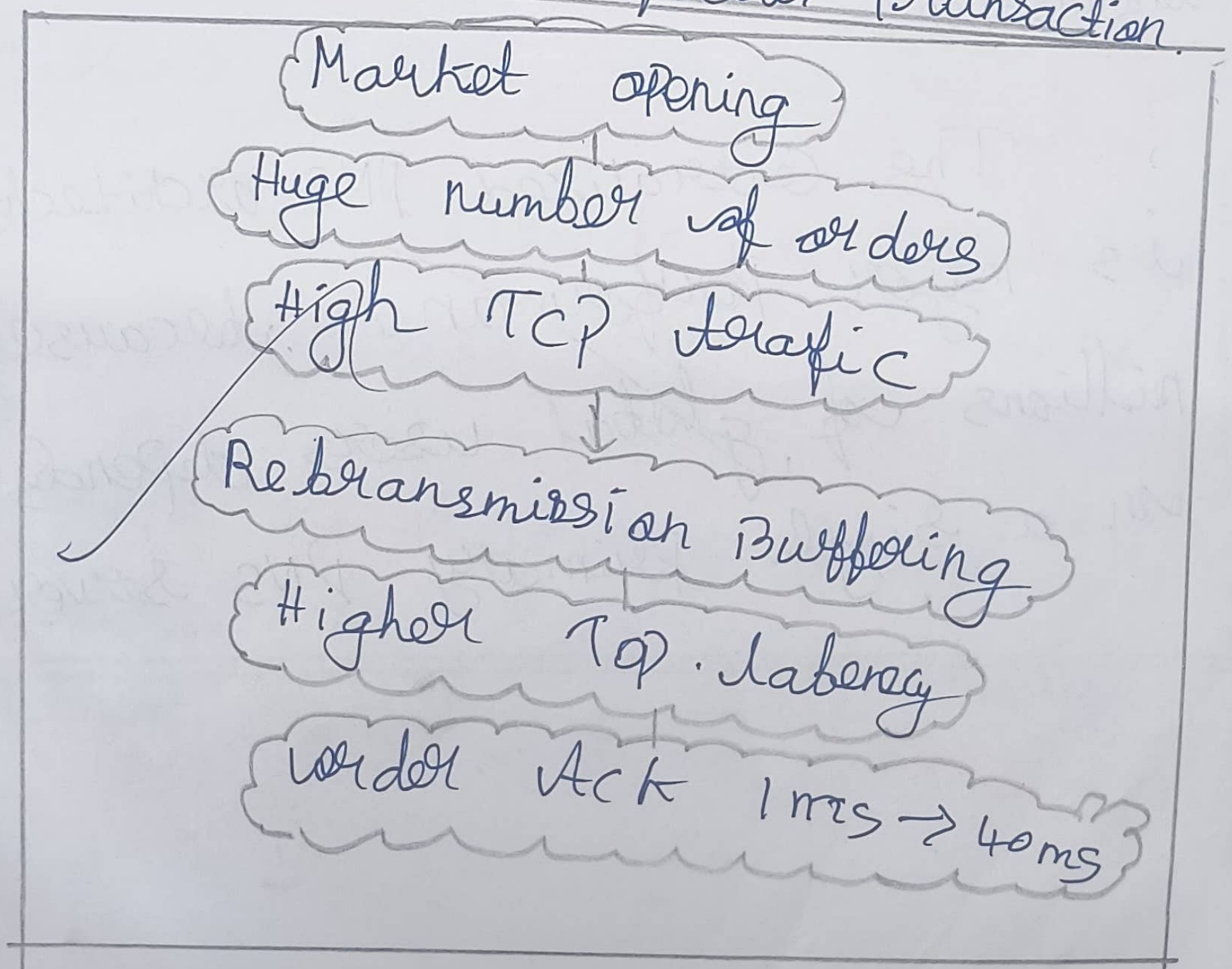
Conclusion :-

The Centralized DNS architecture is major performance because millions of global users depends on a single primary DNS server.

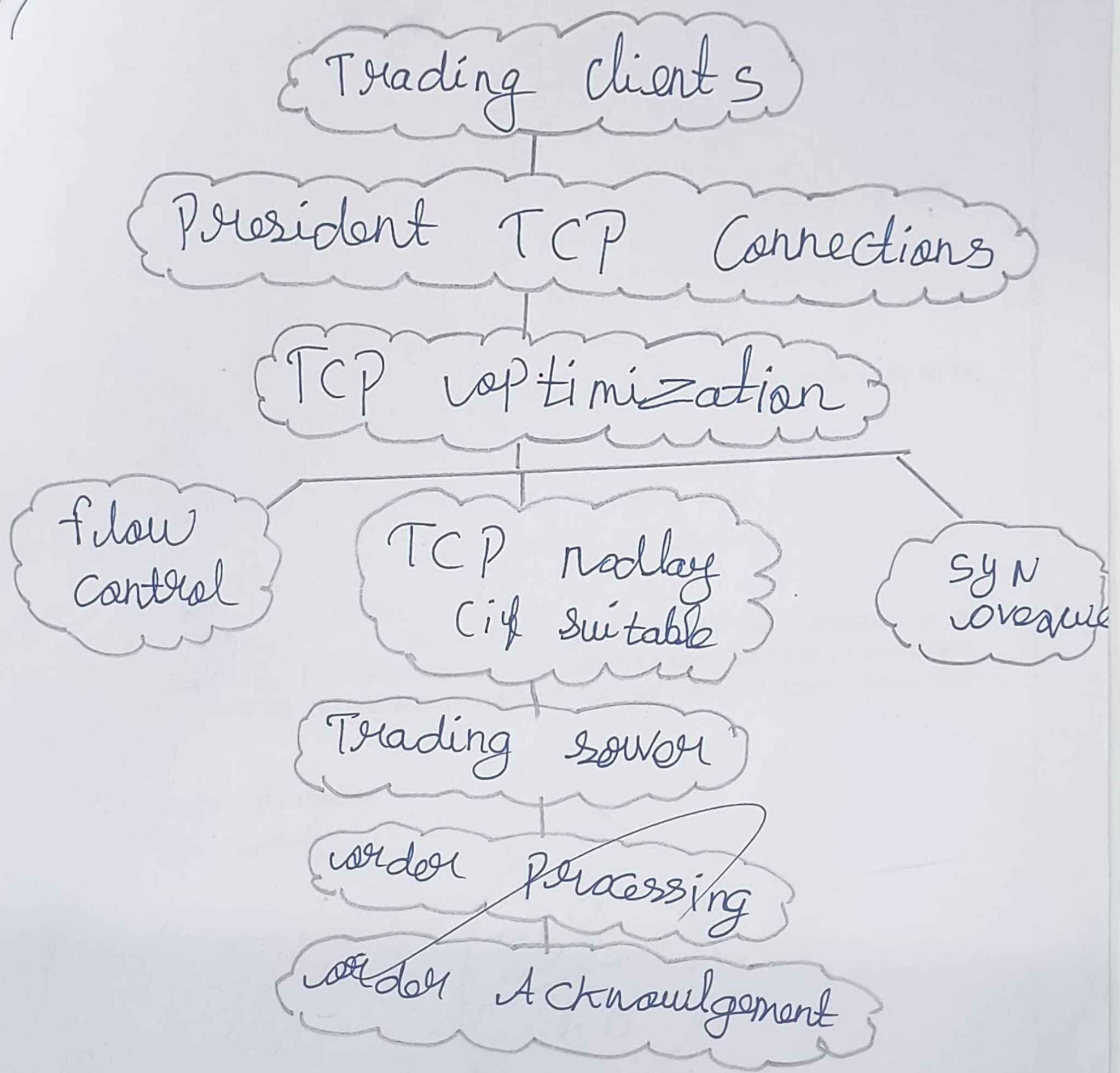
Q4) TCP Latency Analysis in a Electronic Stock

The Financial Institution's Stock Trading Platform handles thousands of buy and sell orders every second and opening hours, acknowledgement latency increases from 1ms to 40ms.

i) Effect of Each factor Transaction



Recommended Architecture:-



Conclusion:

They increase in order of acknowledgment from 1ms to 40ms consistent with combination of TCP control.